rem: random energy model: T_c, frozen 1RSB phase

1 Model

family = rem, variables = ising, disorder = gaussian, constraint = None, prior = None.
random energy model: T_c, frozen 1RSB phase [1, 2]

Hamiltonian and conventions [1, 2]

$$2^N \text{ i.i.d. energies } E_i \sim \mathcal{N}(0, N_s N), \quad N_s = 1/2 \text{ (Derrida's } J = 1\text{; default)} \tag{1}$$

2 Method and route

Route: **replica** (replica method: replicated partition function Z^n, disorder average, order parameters q_{ab} (and conjugates), saddle point, RS / 1RSB ansatz, continuation n $\to$ 0). Rigor: **physics-level**. physics-level: replica $n \to 0$ continuation and RS/1RSB ansatz (Parisi), not proven; numbers carry quadrature error only

3 Derivation

Level (a): each step is rendered from the sympy objects that step put in the State (sympy.latex); nothing is retyped. The level-(a) steps below derive the same model symbolically (`derive_rem.py` checks each step by an independent route and against families/rem.py; `tests/test_rem_pspin.py` runs them with a planted-error control); method as in Derrida, PRL 45, 79 (1980); Phys. Rev. B 24, 2613 (1981).

3.1 Step 1: rem-replicate

Source: [3, 4]. Trace: set `rem_identity`, `rem_tuples`

Replicas 2^N configurations j with `i.i.d.` energies $E_j \sim$ N(0, s N) (s = `N_scale`); n replicas a = 1..n each pick a configuration i_a. The identity below is $(Z^n - 1)/n \to$ ln Z, evaluated by `sympy.limit`; the tuple sum is checked by expanding $(\sum_j w_j)^n$ at n = 2, 3

Replica limit (sympy.limit)

$$\lim_{n \to 0^+} \left(\frac{Z^n - 1}{n} \right) = \log(Z) \tag{2}$$

Replica identity (n $\to$ 0 continuation, physics level)

$$\mathbb{E}(\log(Z)) = \lim_{n \to 0^+} \left(\frac{\log(\mathbb{E}(Z^n))}{n} \right) \tag{3}$$

Partition function

$$Z = \sum_{j=1}^{2^N} e^{-\beta E_j} \tag{4}$$

Replicated partition function: a sum over n-tuples $(i_1, ..., i_n)$

$$Z^n = \sum_{i_1=1}^{2^N} \sum_{i_n=1}^{2^N} e^{-\beta \sum_{a=1}^{n} E_{i_a}} \tag{5}$$

3.2 Step 2: rem-average-disorder

Source: [3, 4]. Trace: set `rem_avg`, `rem_weight`

Disorder average for a fixed tuple the energy sum over the replicas is $\sum_j c_j E_j$ with $c_j = \#\{\text{a: } i_a = \text{j}\}$; the E_j are independent Gaussians, so the log-average is $\sum_j K(-\beta c_j)$, quadratic in c_j, and $\sum_j c_j^2 = \sum_{a,b} q_{ab}$ with $q_{ab} = 1$ if $i_a = i_b$ and 0 otherwise (checked over every tuple of 3 and 4 replicas on 3 configurations)

Gaussian moment generating function (sympy.integrate)

$$\mathbb{E}\left(e^{tE_j}\right) = e^{\frac{Nst^2}{2}} \tag{6}$$

One configuration chosen by c_j replicas

$$\log\left(\mathbb{E}\left(e^{-\beta c_j E_j}\right)\right) = \frac{N\beta^2 c_j^2 s}{2} \tag{7}$$

Fixed tuple: the overlaps q_{ab} in {0, 1}

$$\mathbb{E}\left(e^{-\beta \sum_{a=1}^{n} E_{i_a}}\right) = e^{\frac{N\beta^2 s \sum_{a=1}^{n} \sum_{b=1}^{n} q_{a,b}}{2}} \tag{8}$$

Averaged replicated partition function

$$\mathbb{E}(Z^n) = \sum_{i_1=1}^{2^N} \sum_{i_n=1}^{2^N} e^{\frac{N\beta^2 s \sum_{a=1}^{n} \sum_{b=1}^{n} q_{a,b}}{2}} \tag{9}$$

3.3 Step 3: rem-rs-ansatz

Source: [3, 4]. Trace: set `rem_rsym`

Replica-symmetric ansaetze $q_{ab} = \delta_{ab}$: every replica on its own configuration (2^N $(2^N - 1)$ ... $\sim (2^N)^n$ tuples); $q_{ab} = 1$: all replicas on one configuration (2^N tuples). $\sum_{a,b} q_{ab} = \text{n} +$ `ansatz.sum_offdiag_power`

$q_{ab} = \delta_{ab}$: pair sum

$$\sum_{a=1}^{n} \sum_{b=1}^{n} q_{a,b} = n \tag{10}$$

$q_{ab} = \delta_{ab}$**: (1/N) ln E** Z^n

$$\Phi_n = \frac{\beta^2 n s}{2} + n \log(2) \tag{11}$$

$q_{ab} = \delta_{ab}$**, n** $\to$ **0 (`ansatz.n_to_zero`): the paramagnet**

$$-\beta f = \frac{\beta^2 s}{2} + \log(2) \tag{12}$$

$q_{ab} =$ **1: pair sum**

$$\sum_{a=1}^{n} \sum_{b=1}^{n} q_{a,b} = n^2 \tag{13}$$

$q_{ab} =$ **1: (1/N) ln E** Z^n

$$\Phi_n = \frac{\beta^2 n^2 s}{2} + \log(2) \tag{14}$$

$q_{ab} =$ **1, per replica (`sympy.limit` n** $\to$ **0+ diverges)**

$$\lim_{n \to 0^+} \left(\frac{\Phi_n}{n} \right) = \infty \tag{15}$$

$q_{ab} =$ **1 refused** `ansatz.n_to_zero` refuses it (precondition:polynomial-divisible-by-n: expr = $\beta^2\, n^2\, s/2$ + log(2)): ln 2 / n diverges as n $\to$ 0, so the single-configuration RS saddle is unphysical; the frozen phase needs the block ansatz below

3.4 Step 4: rem-1rsb-blocks

Source: [3, 4]. Trace: set `rem_blocks`

One-step replica symmetry breaking n/m blocks of m replicas; the replicas of a block share one configuration (q1 = 1), different blocks different ones (q0 = 0). The n/m configurations give 2^N $(2^N - 1)$ ... $\sim (2^N)^{n/m}$ tuples; the falling-factorial correction is $O(2^{-}N)$ (shown for 3 blocks, K = 2^N; checked for 2 to 5 blocks) and drops out of (1/N) ln. The pair sum is checked against `ansatz.rsb_matrix` at integer n

Falling-factorial correction, 3 blocks (sympy series in 1/K)

$$\log\left(\frac{(K-2)\,(K-1)}{K^2} \right) = -\frac{3}{K} + O\left(\frac{1}{K^2}; K \to \infty \right) \tag{16}$$

1RSB pair sum (n + `ansatz.sum_offdiag_power`**)**

$$\sum_{a=1}^{n} \sum_{b=1}^{n} q_{a,b} = mn \tag{17}$$

1RSB: (1/N) ln E Z^n

$$\Phi_n = \frac{\beta^2 m n s}{2} + \frac{n \log(2)}{m} \tag{18}$$

3.5 Step 5: rem-replica-limit

Source: [3, 4]. Trace: set `rem_Phim`

Continuation n → 0 Phi(m) = $\lim_{n\to 0}\ \Phi_n$ / n by `ansatz.n_to_zero`; equal (sympy) to `families/rem.py` `rem_1rsb_functional`, and Phi(1) is the $q_{ab} = \delta_{ab}$ paramagnet of step 3

1RSB functional per replica and spin

$$\Phi(m) = \frac{\beta^2 m s}{2} + \frac{\log(2)}{m} \tag{19}$$

3.6 Step 6: rem-extremize

Source: [3, 5]. Trace: set `rem_ext`

Extremization over 0 < m ≤ 1 for n < 1 the 1RSB -beta f is the MINIMUM of Phi(m) over 0 < m ≤ 1 (the n → 0 inversion of the saddle); Phi is convex in m, so the minimum is the stationary point m* when m* ≤ 1 (frozen phase) and m = 1 when m* ≥ 1 (paramagnet). e = -dPhi/dbeta at fixed m (stationarity in m), entropy = Phi + beta e

Stationarity

$$\frac{d}{dm}\Phi(m) = \frac{\beta^2 s}{2} - \frac{\log(2)}{m^2} \tag{20}$$

Stationary block size ($\beta_c = \sqrt{2\ln 2/s}$)

$$m^* = \frac{\sqrt{2}\sqrt{\log(2)}}{\beta\sqrt{s}} = \frac{\beta_c}{\beta} \tag{21}$$

Convexity

$$\frac{d^2}{dm^2}\Phi(m) = \frac{2\log(2)}{m^3} \tag{22}$$

Frozen phase (m* < 1): -beta f = Phi(m*)

$$-\beta f = \sqrt{2}\beta\sqrt{s}\sqrt{\log(2)} \tag{23}$$

Frozen phase: energy, free energy and entropy per spin

$$e = -\sqrt{2}\sqrt{s}\sqrt{\log(2)} \tag{24}$$

$$f = -\sqrt{2}\sqrt{s}\sqrt{\log(2)} \tag{25}$$

$$S = 0 \tag{26}$$

Paramagnet $(m \geq 1, m = 1)$**: -beta f, energy, entropy**

$$-\beta f = \frac{\beta^2 s}{2} + \log(2) \tag{27}$$

$$e = -\beta s \tag{28}$$

$$S = -\frac{\beta^2 s}{2} + \log(2) \tag{29}$$

At the requested beta branch: paramagnet, m = 1.00000000000000

3.7 Step 7: rem-derrida-check

Source: [3, 1, 2]. Trace: set `rem_derrida`

Derrida's direct solution microcanonical entropy (`families/rem.py` `rem_entropy`): its negative root e0 is the ground-state energy and β_c = s'(e0) (the slope where the entropy vanishes). sympy checks both equal the replica β_c and the frozen f of step 6

Microcanonical entropy

$$s(e) = -\frac{e^2}{2s} + \log(2) \tag{30}$$

Ground-state energy and critical inverse temperature

$$e_0 = -\sqrt{2}\sqrt{s}\sqrt{\log(2)} \tag{31}$$

$$\beta_c = \frac{\sqrt{2}\sqrt{\log(2)}}{\sqrt{s}} \tag{32}$$

Agreement at the requested parameters beta = 1.0, `N_scale` = 0.5: branch paramagnet, m = 1.0; replica f = -0.94314718055994530942, Derrida `rem_free_energy` = -0.94314718055994530942, |difference| = 1.97e-31 (30-digit evaluation, tolerance 1e-25)

3.8 Equations implemented by the family code

Conventions e = E/N, N_s = s (energy variance per spin); entropy, β_c and the free energy as in `families/rem.py`

Microcanonical entropy [1, 2] (sympy form; checked against the code by `tests/test_latex.py`)

$$s(e) = -\frac{e^2}{2s} + \log(2) \tag{33}$$

Critical inverse temperature $(s(e_0) = 0, \beta_c = s'(e_0))$ (sympy form; checked against the code by `tests/test_latex.py`)

$$\beta_c = \frac{\sqrt{2}\sqrt{\log(2)}}{\sqrt{s}} \tag{34}$$

Paramagnet, beta $\leq \beta_c$ (annealed = quenched) (sympy form; checked against the code by `tests/test_latex.py`)

$$-\beta f = \frac{\beta^2 s}{2} + \log(2) \tag{35}$$

1RSB functional (blocks of m replicas share a configuration; n $\to$ 0) (sympy form; checked against the code by `tests/test_latex.py`)

$$-\beta f = \frac{\beta^2 m^2 s + \log(4)}{2m} \tag{36}$$

Frozen phase, beta $> \beta_c$ ($m = \beta_c/\beta$) (sympy form; checked against the code by `tests/test_latex.py`)

$$f = -\sqrt{2}\sqrt{s}\sqrt{\log(2)} \tag{37}$$

4 Calculation at the requested parameters

The run's trace, in order (skipped steps say why).

4.1 Step 1: rem-critical-temperature

Source: [1, 2]. Trace: set errors, `step_note`, values; computed `N_scale`, β_c, T_c

4.2 Step 2: rem-saddle

Source: [1, 2, 6]. Trace: set errors, `step_note`, values; paramagnet ($\beta \leq \beta_c$): annealed = quenched, -beta f = ln 2 + `N_scale` $\beta^2/2$

5 Solution

Values at the requested parameters (beta = 1.0), each with its finality: only stable and exact-by-literature values are final.

key	value	finality (ansatz)	final
beta_c	1.6651092223154	exact-by-literature (rs) [7, 2, 5]	yes
T_c	0.600561204393225	exact-by-literature (rs) [7, 2, 5]	yes
f	-0.943147180559945	exact-by-literature (rs) [7, 2, 5]	yes
q	0.0	exact-by-literature (rs) [7, 2, 5]	yes

Other outputs (parameters, labels, diagnostics): `N_scale` = 0.5; ansatz = rs; stable = None.

6 Stability

Computed stability quantities (replicon / AT / cavity susceptibility): stable = None.

exact-by-literature: β_c, T_c, f, q.

7 Reference check

Banked values (reference/*.json) against this pipeline's output; numbers are shown to 16 significant digits, the comparison uses them in full; slow and statistical rows run only with –full (`render_latex(slow=True)`).

key	computed	banked	tolerance	verdict	source
tc_derrida	0.6005612043932249	0.6005612043932249	1e-20	agrees	[7, 8, 9]
ground_energy_derrida	-0.8325546111576978	-0.8325546111576978	1e-20	agrees	[7, 10]
m_times_beta_over_beta_c	1.0	1	1e-20	agrees	[11, 12]

8 Labels

Rigor: **physics-level**. Route: **replica**.

physics-level: replica $n \to 0$ continuation and RS/1RSB ansatz (Parisi), not proven; numbers carry quadrature error only

References

[1] Derrida, PRL 45, 79 (1980)

[2] Derrida, Phys. Rev. B 24, 2613 (1981) (exact solution: microcanonical entropy s(e))

[3] replicant level (a) derivation (this package)

[4] Mezard, Parisi & Virasoro, Spin Glass Theory and Beyond (1987)

[5] Derrida & Mottishaw, arXiv:1410.1432 (m = T/T_c)

[6] 1RSB form: Mezard, Parisi & Virasoro (1987)

[7] Derrida, The random energy model, Physics Reports 67, 29-35 (1980), Sec. 4: 'T_c = J/2 sqrt(log 2)' with P(E) = (pi N J^2)^(-1/2) exp(-E^2/(N J^2)), Eq. (2)

[8] read from the copy at thp.uni-koeln.de/krug/teaching-Dateien/WS2011/Derrida1980.pdf. Same statement in Derrida & Mottishaw, J. Stat. Mech. (2015) P01021, doi:10.1088/1742-5468/2015/01/P01021, arXiv:1410.1432, Eq. (2). Full paper Derrida, Phys. Rev. B 24, 2613 (1981), doi:10.1103/PhysRevB.24.2613 (not fetched). The paper states the closed form

[9] value = its decimal at J = 1 (replicant N_scale = J^2/2 = 1/2)

[10] also Derrida & Mottishaw, arXiv:1410.1432, Eq. (3). Closed form, decimal at J = 1. In the frozen phase f = E_ground / N (entropy 0)

[11] Derrida & Mottishaw, J. Stat. Mech. (2015) P01021, arXiv:1410.1432, Eqs. (4)-(6): Parisi q(x) = Theta(x - 1 + <P2>) with <P2> = 1 - mu, mu = T/T_c below T_c, so the 1RSB breakpoint is m = T/T_c = beta_c/beta

[12] same in Derrida & Mottishaw, J. Phys. A 55 (2022), doi:10.1088/1751-8121/ac71e9, arXiv:2202.12584, Eq. (5) (<Y_2> = 1 - beta_c/beta). Meaning: m * beta / beta_c = 1 for beta >= beta_c